

服务器准备

IP	配置	安装软件	作用
172.16.5.30	2C2G	Java,maven	运行ruoyi 后端 java 服务
172.16.5.30	2C2G	Mysql,redis	运行 ruoyi 数据库
172.16.5.30	2C2G	Nginx	运行 ruoyi 前端页面和后端反向代理

环境准备

⚠ Caution

服务器 1: 172.16.5.30 (部署 ruoyi 前端项目服务器)

在 172.16.5.51 服务器上

安装 java 运行环境

```
$ dnf install java-17-openjdk java-17-openjdk-devel -y
```

安装 maven 编译工具

```
$ dnf install maven-openjdk17 -y
```

检查安装结果

```
$ mvn -version
Apache Maven 3.6.3 (Red Hat 3.6.3-22)
Maven home: /usr/share/maven
Java version: 17.0.15, vendor: Red Hat, Inc., runtime:
/usr/lib/jvm/java-17-openjdk-17.0.15.0.6-3.el9.alma.1.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "5.14.0-570.24.1.el9_6.x86_64", arch:
"amd64", family: "unix"

$ java -version
openjdk version "17.0.15" 2025-04-15 LTS
OpenJDK Runtime Environment (Red_Hat-17.0.15.0.6-1) (build 17.0.15+6-
LTS)
OpenJDK 64-Bit Server VM (Red_Hat-17.0.15.0.6-1) (build 17.0.15+6-
LTS, mixed mode, sharing)
```

Mysql的安装和配置

⚠ Caution

服务器 IP: 172.16.5.31

安装MySQL

```
$ dnf install mysql mysql-server

$ systemctl enable --now mysqld
```

配置MySQL

设置 root 密码

```
$ mysql_secure_installation
```

```
[root@nginx-03 ~]# mysql_secure_installation
```

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.

VALIDATE PASSWORD COMPONENT can be used to test passwords and improve security. It checks the strength of password and allows the users to set only those passwords which are secure enough. Would you like to setup VALIDATE PASSWORD component?

Press y|Y for Yes, any other key for No:

There are three levels of password validation policy:

LOW Length >= 8

MEDIUM Length >= 8, numeric, mixed case, and special characters

STRONG Length >= 8, numeric, mixed case, special characters and dictionary

Please enter 0 = LOW, 1 = MEDIUM and 2 = STRONG:

Please set the password for root here.

New password: **abcd1234**

Re-enter new password: **abcd1234**

Remove anonymous users? (Press y|Y for Yes, any other key for No) :
Success.

Normally, root should only be allowed to connect from 'localhost'. This ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? (Press y|Y for Yes, any other key for No) :
Success.

By default, MySQL comes with a database named 'test' that anyone can access. This is also intended only for testing, and should be removed before moving into a production environment.

Remove test database and access to it? (Press y|Y for Yes, any other key for No) :
- Dropping test database...
Success.

- Removing privileges on test database...
Success.

Reloading the privilege tables will ensure that all changes made so far will take effect immediately.

Reload privilege tables now? (Press y|Y for Yes, any other key for No) :
Success.

All done!

验证一下

```
$ mysql -uroot -pabcd1234
```

```
[root@nginx-03 ~]# mysql -uroot -pabcd1234
mysql: [Warning] Using a password on the command line interface can be insecure
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.41 Source distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql>  看到这个标识说明登录成功
```

退出登陆

```
mysql> exit;
```

修改监听地址

```
/etc/my.cnf.d/mysql-server.cnf
```

注意：写自己的服务器 IP 地址

```
[mysqld]
bind-address=172.16.5.31
```

重启服务

```
$ systemctl restart mysqld
```

导入数据

数据库文件存放目录

```
$ wget https://gitee.com/y_project/RuoYi-Vue/repository/archive/v3.8.9.zip
$ unzip v3.8.9.zip
$ cd RuoYi-Vue-v3.8.9/sql
-rw-r--r-- 1 root root 11985 Dec 30 2024 quartz.sql
-rw-r--r-- 1 root root 57227 Dec 30 2024 ry_20240629.sql
```

创建 ruoyi 用户，并设置访问权限

```
# 登录数据库
$ mysql -uroot -p12345678
```

```
mysql> create database ruoyidb;
mysql> create user 'ruoyi'@'%' identified by 'Ruoyi.123';
mysql> grant all on `ruoyidb`.* to 'ruoyi'@'%';
mysql> show grants for 'ruoyi'@'%';
mysql> exit;
```

导入数据

```
# 不登陆数据库
cd /home/codes/RuoYi-Vue-v3.8.9/sql/
mysql -uruoyi -pRuoyi.123 ruoyidb < ry_20240629.sql
mysql -uruoyi -pRuoyi.123 ruoyidb < quartz.sql

# 登录数据库
mysql > use ruoyidb;
mysql > source /home/codes/RuoYi-Vue-v3.8.9/sql/ry_20240629.sql;
```

Redis的安装和配置

安装Redis

```
$ dnf install redis
```

配置redis

```
$ mkdir -p /data/redis/6379  
$ chown redis.redis /data/redis/6379/ -R  
$ vim /etc/redis/redis.conf
```

```
bind 0.0.0.0  
daemonize no  
supervised no  
requirepass 123456  
masterauth 123456  
protected-mode yes  
port 6379  
dir /data/redis/6379  
pidfile /data/redis/6379/redis_6379.pid  
logfile "/data/redis/6379/redis_6379.log"  
tcp-backlog 511  
timeout 0  
tcp-keepalive 300  
loglevel notice  
databases 16  
always-show-logo no  
set-proc-title yes  
proc-title-template "{title} {listen-addr} {server-mode}"  
stop-writes-on-bgsave-error yes  
rdbcompression yes  
rdbchecksum yes  
dbfilename dump.rdb  
rdb-del-sync-files no  
replica-serve-stale-data yes  
replica-read-only yes  
repl-diskless-sync no  
repl-diskless-sync-delay 5  
repl-diskless-load disabled  
repl-disable-tcp-nodelay no  
replica-priority 100
```

```
acllog-max-len 128
lazyfree-lazy-eviction no
lazyfree-lazy-expire no
lazyfree-lazy-server-del no
replica-lazy-flush no
lazyfree-lazy-user-del no
lazyfree-lazy-user-flush no
oom-score-adj no
oom-score-adj-values 0 200 800
disable-thp yes

no-appendfsync-on-rewrite no
auto-aof-rewrite-percentage 100
auto-aof-rewrite-min-size 64mb
aof-load-truncated yes
aof-use-rdb-preamble yes
lua-time-limit 5000
slowlog-log-slower-than 10000
slowlog-max-len 128
latency-monitor-threshold 0
notify-keyspace-events ""
hash-max-ziplist-entries 512
hash-max-ziplist-value 64
list-max-ziplist-size -2
list-compress-depth 0
set-max-intset-entries 512
zset-max-ziplist-entries 128
zset-max-ziplist-value 64
hll-sparse-max-bytes 3000
stream-node-max-bytes 4096
stream-node-max-entries 100
activerehashing yes
client-output-buffer-limit normal 0 0 0
client-output-buffer-limit replica 256mb 64mb 60
client-output-buffer-limit pubsub 32mb 8mb 60
hz 10
dynamic-hz yes
aof-rewrite-incremental-fsync yes
rdb-save-incremental-fsync yes
jemalloc-bg-thread yes

save 3600 1 # 3600秒（1小时）内至少有1个key被修改
```

```
save 300 100 # 300秒（5分钟）内至少有100个key被修改
save 60 10000 # 60秒内至少有10000个key被修改
```

修改内核参数

```
$ echo "vm.overcommit_memory = 1" >> /etc/sysctl.conf
$ echo "net.core.somaxconn = 1024" >> /etc/sysctl.conf
$ sysctl -p
```

启动服务

```
$ systemctl daemon-reload
$ systemctl enable --now redis
```

部署后端服务

⚠ Caution

服务器 IP: 172.16.5.30

编译后端服务

```
$ cd /home/RuoYi-Vue-v3.8.9/
$ mvn clean package -Dmaven.test.skip=true
$ ls ruoyi-admin/target/ruoyi-admin.jar
$ mkdir -p /app/ruoyi/config
$ cp ruoyi-admin/target/ruoyi-admin.jar /app/ruoyi
$ cp ruoyi-admin/src/main/resources/*.yml /app/ruoyi/config/
```

修改配置

```
$ cd /app/ruoyi/config
$ vim application.yml
```



```
# redis 配置
redis:
  # 地址
  host: localhost
  # 端口，默认为 6379
  port: 6379
  # 数据库索引
  database: 0
  # 密码
  password:
  # 连接超时时间
  timeout: 10s
  lettuce:
    pool:
      # 连接池中的最小空闲连接
```

```
$ vim application-druid.yml
```

```
# 数据源配置
spring:
  datasource:
    type: com.alibaba.druid.pool.DruidDataSource
    driverClassName: com.mysql.cj.jdbc.Driver
    druid:
      # 主库数据源
      master:
        url: jdbc:mysql://localhost:3306/ry-vue?useUnicode=true&characterEncoding=utf8&useSSL=true&serverTimezone=GMT%2B8
        username: root
        password: password
      # 从库数据源
      slave:
        # 从数据源开关/默认关闭
```

启动服务

```
(( "java", pid=39295, fd=22))
```

修改Nginx配置

```
$ vi /etc/nginx/conf.d/ruoyi.conf

# 添加location
location /prod-api/ {
    proxy_set_header Host $http_host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header REMOTE-HOST $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_pass http://localhost:8080/;
}

$ nginx -t
$ systemctl restart nginx
```

附件

```
#!/bin/bash

JAR_DIR="/app/ruoyi"
JAR_NAME="ruoyi-admin.jar"

function getPid() {
    pid=$(ps -ef |grep ${JAR_NAME} |grep -v grep | awk '{print $2}')
    echo $pid
}

function start() {
    pid=$(getPid)
    if [[ -z "$pid" ]]; then
        cd ${JAR_DIR}
        nohup java -jar ruoyi-admin.jar --
spring.config.location=config/ > /dev/null 2>&1 &
        echo "starting..."
    else
        echo "process is running,pid:${pid}"
    fi
}

function stop() {
```

```

pid=$(getPid)
if [[ -z "$pid" ]]; then
    echo "process is not exists"
else
    kill -9 $pid
    echo "stop pid:${pid}"
fi
}

case "$1" in
    start)
        start
        ;;
    stop)
        stop
        ;;
    restart)
        stop
        start
        ;;
    status)
        pid=$(getPid)
        if [[ -z "$pid" ]]; then
            echo "进程不存在"
        else
            echo "process id is $pid"
        fi
        ;;
    *)
        echo "params: start/stop/restart/status"
esac

```